

INTRODUCTION

The Fixed Inclinator System is designed for automated monitoring of deep horizontal displacement below the ground surface. It is widely used in landslide prevention, construction monitoring, dam safety monitoring, tunnel convergence, and other geotechnical engineering scenarios.

The inclinometer casing is installed vertically underground. Ground movement causes displacement and deformation of the casing, while the fixed inclinometer probes installed inside the casing collect data automatically at configured intervals to track cumulative deformation over time.



OVERVIEW

FIXED INCLINOMETER SYSTEM

Fixed Inclinator



Controller



Probe



Measured Data

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● 2020-01-10 09:46:38

● 2020-01-09 12:46:51

● 2019-12-23 14:03:39

● 2019-12-21 13:38:11

● 2019-12-14 08:25:55

● 2019-12-13 09:59:22

● 2019-12-12 10:05:44

● 2019-12-11 10:03:09

● 2019-12-11 11:55:51

● 2019-12-09 15:21:40

● 2019-12-08 10:26:40

● 2019-12-06 10:18:36

● 2019-12-01 14:33:42

● 2019-11-24 13:20:57

● 2019-11-22 09:22:17

● 2019-11-19 10:13:39

● 2019-11-16 14:50:29

● 2019-11-14 10:05:31

● 2019-11-12 14:26:15

● 2019-11-10 09:01:52

● 2019-11-07 14:51:37

● 2019-11-06 14:44:07

● 2019-11-06 10:38:32

● 2019-11-03 09:52:26

● 2019-11-28 10:24:25

● 2019-11-05 14:53:09

● 2019-11-03 09:52:25

● 2019-10-27 11:03:09

● 2019-10-25 14:39:21

● 2019-10-22 14:16:54

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AI-Powered, Real-Time Automation

Unattended Operation, Safe & Reusable

High-Precision 3-Axis Sensing, Smart Curve Generation

Temperature-Compensated & Waterproof to 200 m

KEY FEATURES

AI-DRIVEN, REAL-TIME AUTOMATION

Seamlessly integrates data acquisition systems with AI analysis software for effortless, 24/7 automated monitoring and data processing.

HIGH-PRECISION 3-AXIS, SMART PROFILING

Features industrial-grade 3-axis sensors that automatically generate clear vertical or horizontal displacement curves, making deformation trends instantly visible.

UNATTENDED OPERATION, HIGHLY REUSABLE

Requires zero manual intervention during operation. Fully reusable across multiple deployments, providing cost-effective safety guarantees for critical projects.

TEMP-COMPENSATED, 2MPA WATERPROOF

Built-in temperature compensation algorithms and exceptional waterproofing up to 2MPa (equivalent to 200 meters of water depth) ensure reliable performance in the harshest environments.

APPLICATIONS



Geological Landslide



Construction Engineering Construction Monitoring



Dam Monitoring

TECHNICAL SPECIFICATIONS

Parameter	Specification
Function type	Fixed inclinometer system
Product model	RIP1025S
System net weight	1 kg
Sensor quantity	MEMS × 1
Rod length	0.79 m
Positioning wheel spacing	0.5 m - 1 m
Measurement range	±15° / ±30°
Resolution	0.01 mm
Repeatability	±0.003°
Operating temperature	-20°C~60°C
Main materials	Carbon fiber and aluminum alloy
Tensile performance	≥5.5 kN (550 kgf)
Waterproof sealing	2MPa (200 m underwater)
Monitoring frequency	Up to once per minute
Data storage	Offline data is automatically stored in the collector; onboard storage ≥ 16G; supports more than 1 year of historical data
Data output requirement	Direct output of spatial coordinate values (X, Y, Z)
Wireless communication	4G/5G full-network support, supports user SIM card
Network requirement	Supports APN network; data returns directly and must not be exposed to the public network
Wired communication	RS-485
Communication protocols	HTTP protocol, MQTT protocol
Network transparent transmission	Supported; supports two or more sockets
Structure	Segmented assembly structure; length can be spliced on site
Single section length	1 m
Connection cable	One cable leads the complete device to the collector end
Torsion correction	Supported
Offset correction	Supported
Convergence monitoring	Supports tunnel convergence monitoring requirements
Remote device management	Device status monitoring, diagnosis, remote configuration, remote upg rade, remote registration
Power supply	DC12V
Power consumption	≤0.05W / section